

2 Differentiation

3 Differentiation II

4 Differentiation III

Differentiation



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 3A03 Real Analysis I

Instructor: David Earn

Lecture 2
Differentiation
Wednesday 8 January 2025

Survey to do right now

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Announcements

Announcements

- Results of Survey 1

Announcements

- Results of Survey 1
- Results of Survey 2

Background / reminder

Definition (Cauchy sequence)

Background / reminder

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A sequence $\{s_n\}$

Background / reminder

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A sequence $\{s_n\}$ is said to be

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A sequence $\{s_n\}$ is said to be a *Cauchy sequence*

Background / reminder

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A sequence $\{s_n\}$ is said to be a *Cauchy sequence* iff

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Background / reminder

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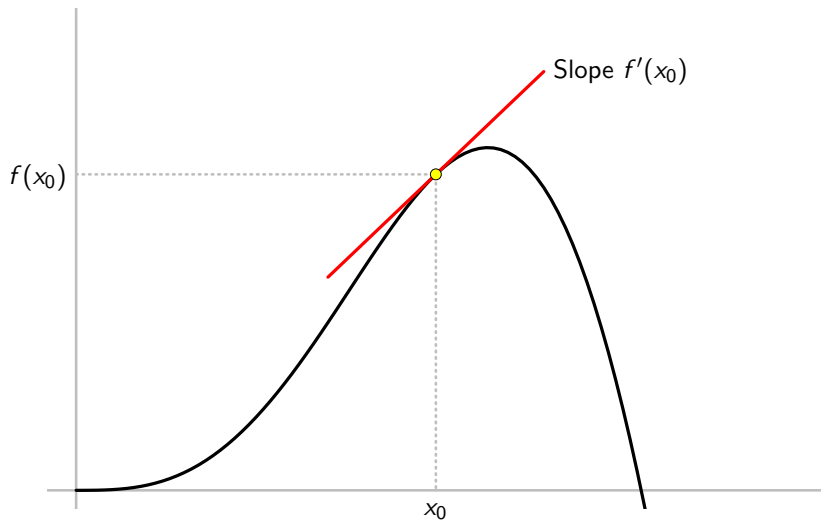
A sequence $\{s_n\}$ is said to be a *Cauchy sequence* iff for all $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that if $m \geq N$ and $n \geq N$ then $|s_n - s_m| < \varepsilon$.

Poll: another background check

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- .

The Derivative

The Derivative



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Definition (Derivative)

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Let f be defined on an interval I

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- Two things are equal, but in one $x \neq 2$ and in the other $x = 2$.
- Good illustration of why it is important to define the meaning of limits rigorously.

Poll

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Let f be defined in a neighbourhood I of 0

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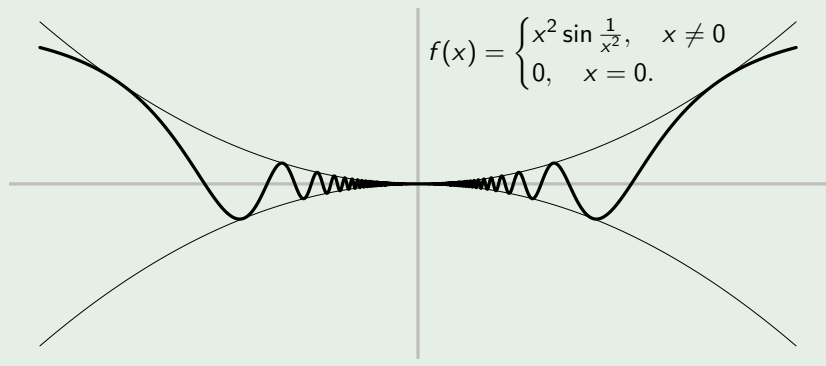
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$$\text{Suppose } f(x) = \begin{cases} x^2 \sin \frac{1}{x^2}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

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$\therefore f$ is differentiable at 0 and $f'(0) = 0$. □

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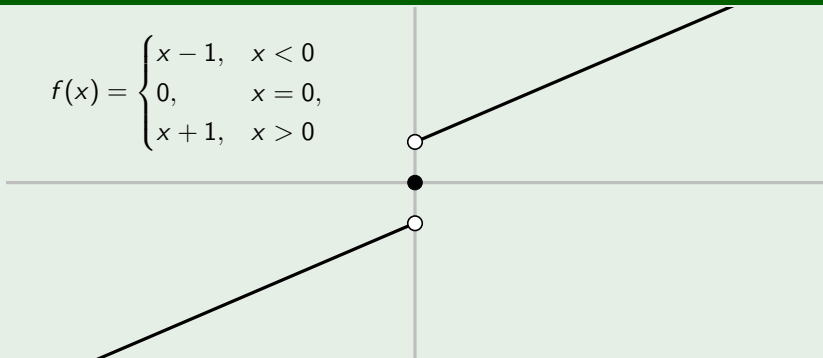
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Note: If x_0 is not an endpoint of the interval I then f is differentiable at x_0 iff $f'_+(x_0) = f'_-(x_0) \neq \pm\infty$.

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Example

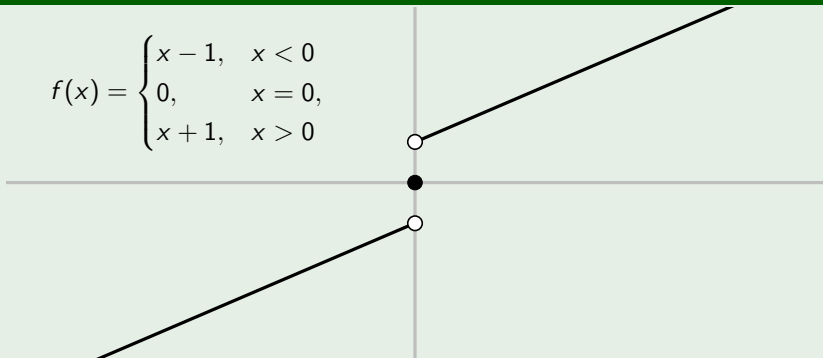
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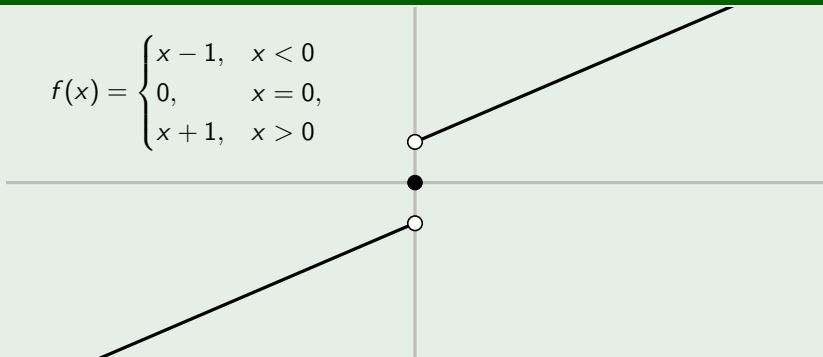


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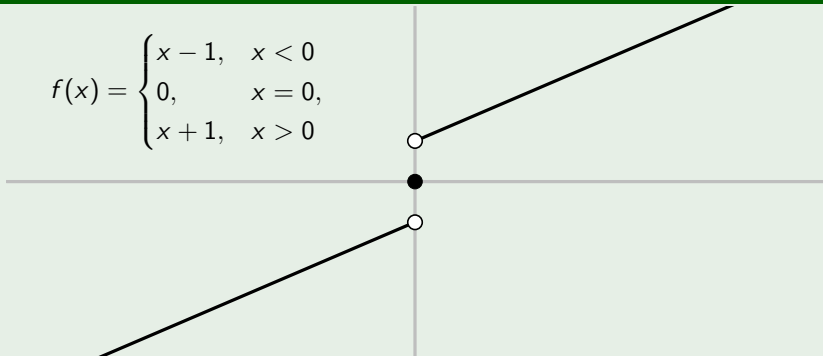


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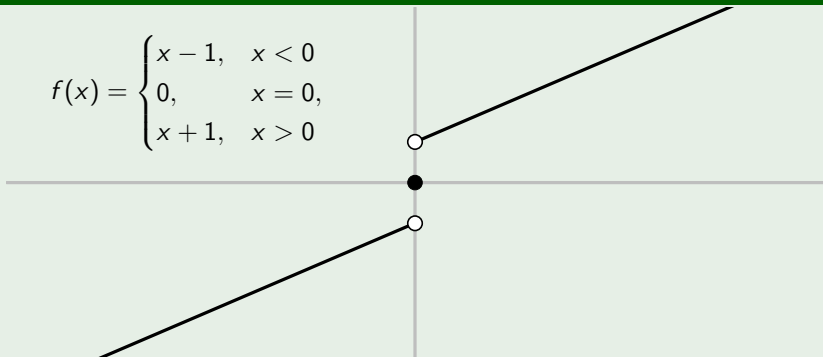


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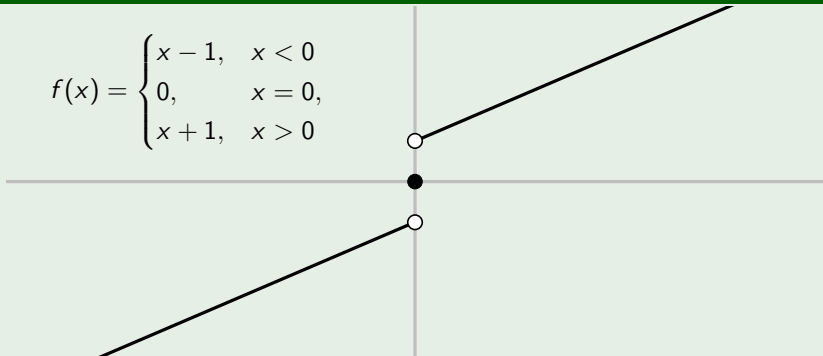


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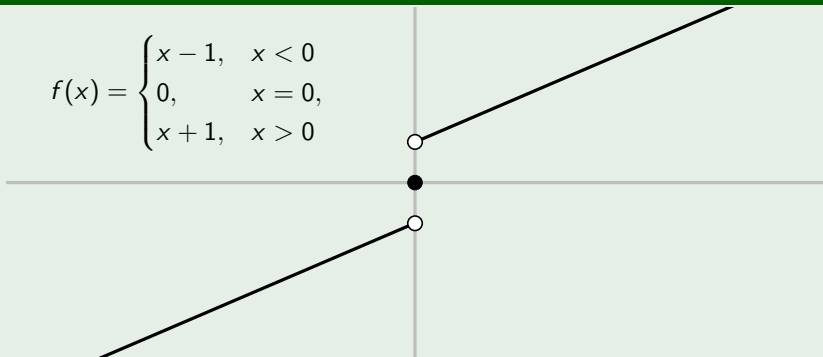


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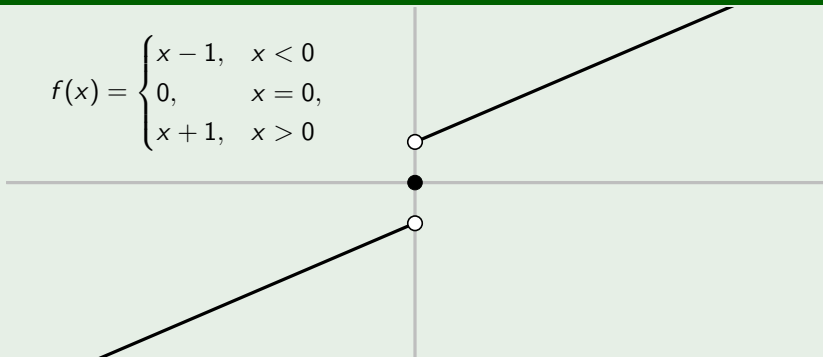


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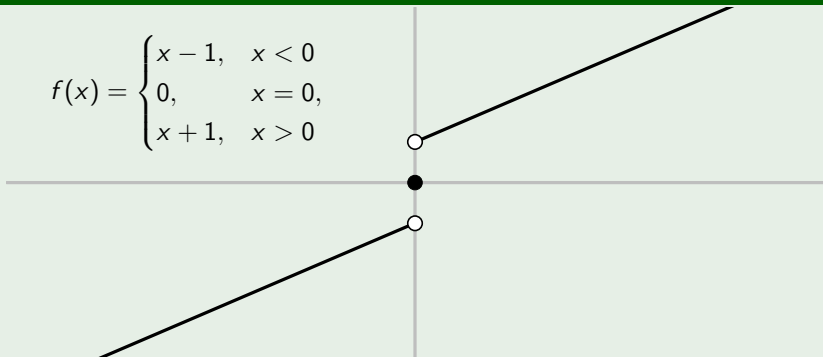


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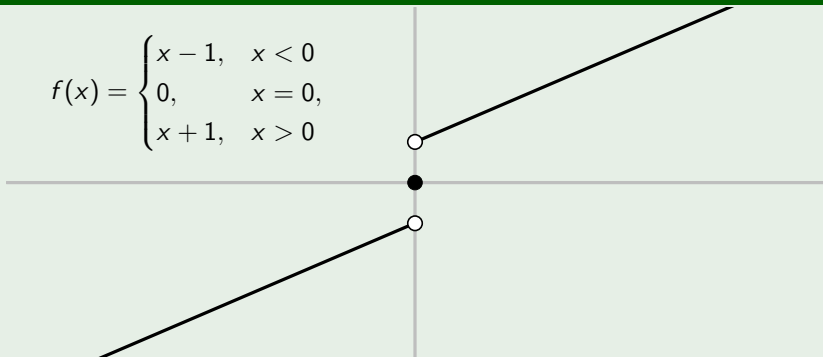


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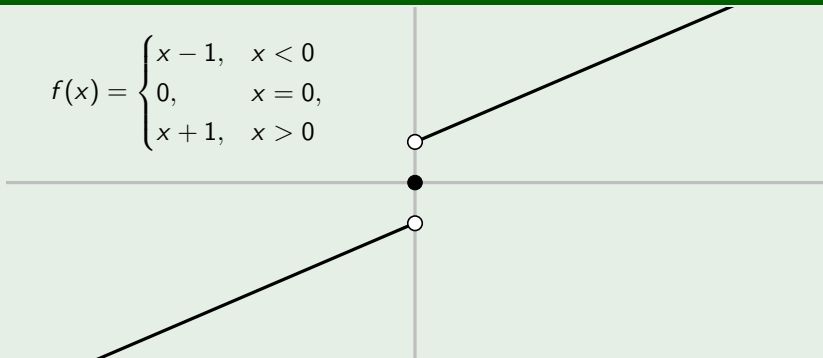


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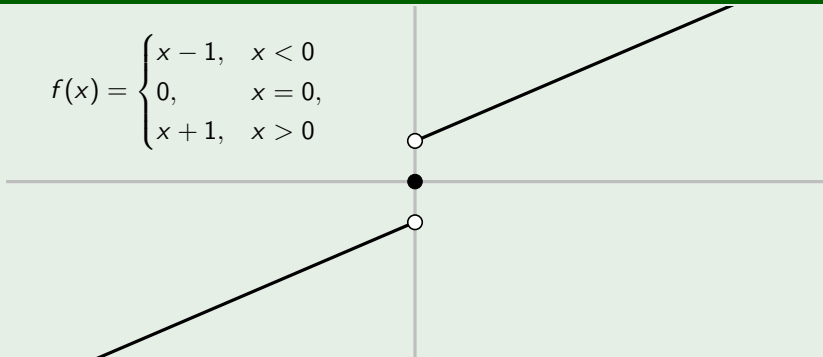


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(TBB §2.7, and problem 2.7.4)

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where we have used the theorem on the algebra of limits. □



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 3A03 Real Analysis I

Instructor: David Earn

Lecture 3
Differentiation II
Friday 10 January 2025

Announcements

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- Course evaluation scheme has changed (next two slides).
 - The [course web site](#) and [online syllabus](#) have not yet been updated to reflect these changes, but that will happen soon (hopefully over the weekend).

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Last time...

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- Definition of the derivative.

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 - Example: Trapping Principle

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- Proved differentiable $\implies$ continuous.

More on the derivative

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Theorem (Algebra of derivatives)

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Suppose f and g are defined on an interval I and $x_0 \in I$.

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- 4** $\left(\frac{f}{g}\right)'(x_0) = \left(\frac{gf' - fg'}{g^2}\right)(x_0) \quad (g(x_0) \neq 0).$

(TBB Theorem 7.7, p. 408)

The Derivative

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Theorem (Chain rule)

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Suppose f is defined in a neighbourhood U of x_0

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(TBB §7.3.2, p. 411)

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- Can we use $(\spadesuit)$ to prove the chain rule?

Poll

- Go to
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- Click on **Math 3A03**
- Click on **Take Class Poll**
- Fill in poll **Derivatives: Chain Rule**
- .

REMINDER: limits of functions

Theorem (Equivalence of ε - δ and sequence definitions of limits)

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Note: The deleted neighbourhood $(I \setminus \{x_0\})$ can be replaced by any set on which f is defined and x_0 is an accumulation point.

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Note: TBB's proof leaves out the proof that $f'(x_0) = 0$ in case 2 above. 

More on the derivative

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Theorem (Derivative at local extrema)

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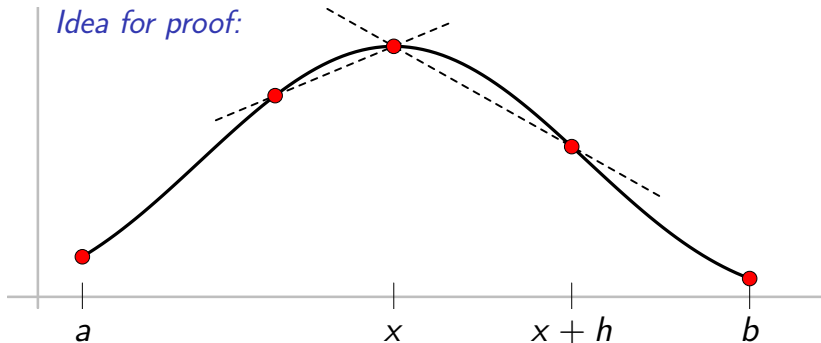
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Idea for proof:



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If f has a local maximum

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But since f is differentiable at x , the left and right derivatives must be equal, hence $f'(x) = 0$. □



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 3A03 Real Analysis I

Instructor: David Earn

Lecture 4
Differentiation III
Monday 13 January 2025

Poll

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 3A03**
- Click on **Take Class Poll**
- Fill in poll **Office hours**
- .

Announcements

Announcements

- In-Class polls:

Announcements

- In-Class polls: if you do not have a device that enables you to participate in polls in class,

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Announcements

- In-Class polls: if you do not have a device that enables you to participate in polls in class, or if any other issue prevents you from participating in polls, please let me know by e-mail.
- This Friday (17 Jan 2025), I will be out of town and the class will be a Q&A session with the TA.

Announcements

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- The [online syllabus](#) has been revised to account for the changes in how the course will be evaluated (see [slides from Lecture 3](#)).

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Students may use generative AI throughout this course in whatever way enhances their learning; no special documentation or citation is required. Note that access to generative AI will **not** be available during tests or exams.

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- The [course web site](#) has been updated to reflect the changes in how the course will be evaluated.

Assignment 1

Assignment 1

- [Assignment 1](#) has been posted on the course web site.

Last time...

Last time...

- Discussed algebra of derivatives and chain rule.

Last time...

- Discussed algebra of derivatives and chain rule.
- Proved the chain rule.

Last time...

- Discussed algebra of derivatives and chain rule.
- Proved the chain rule.
- Proved that that derivative is zero at extrema.

The Mean Value Theorem

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Theorem (Rolle's theorem)

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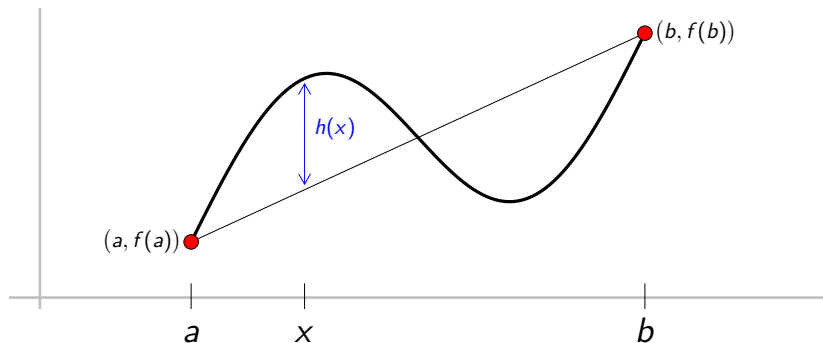
The Mean Value Theorem

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Idea for proof:

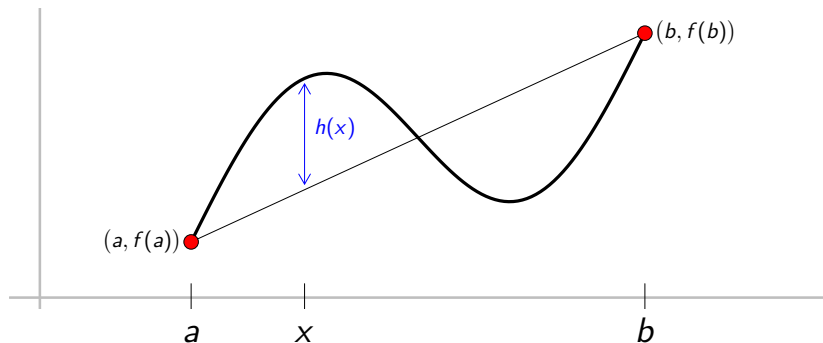
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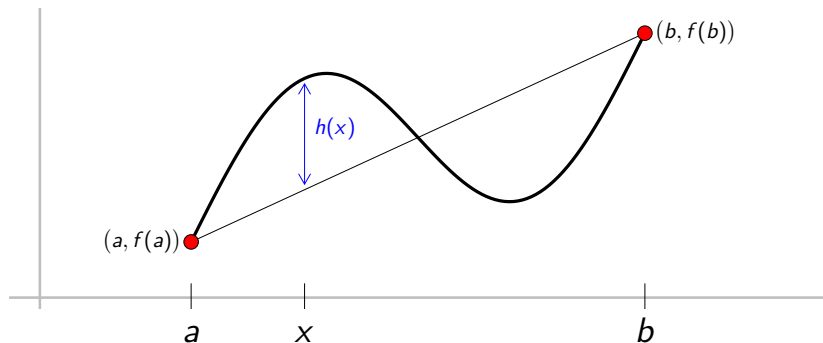
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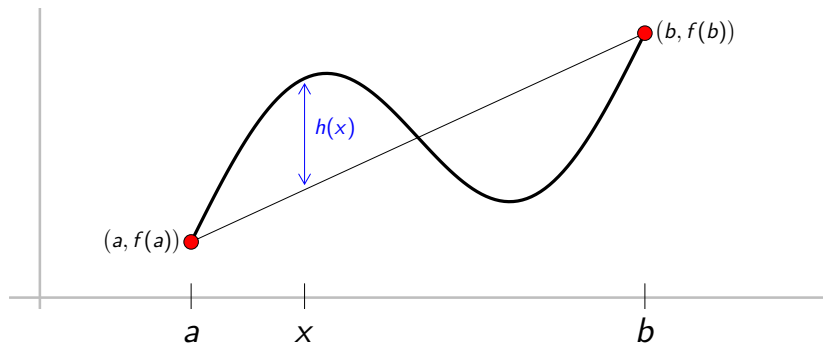


Proof.

Apply Rolle's theorem to

The Mean Value Theorem

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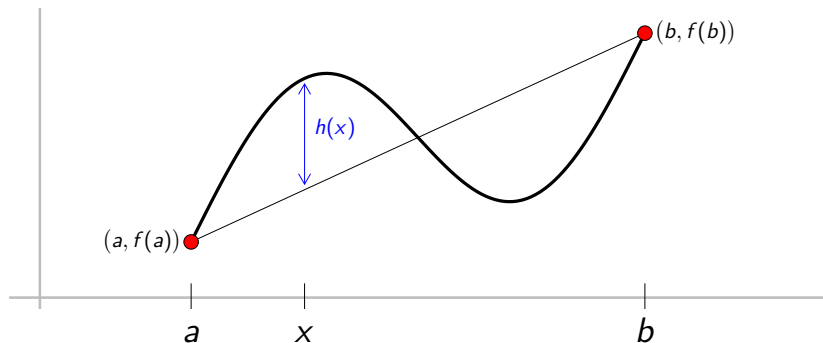
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Apply Rolle's theorem to

$$h(x)$$

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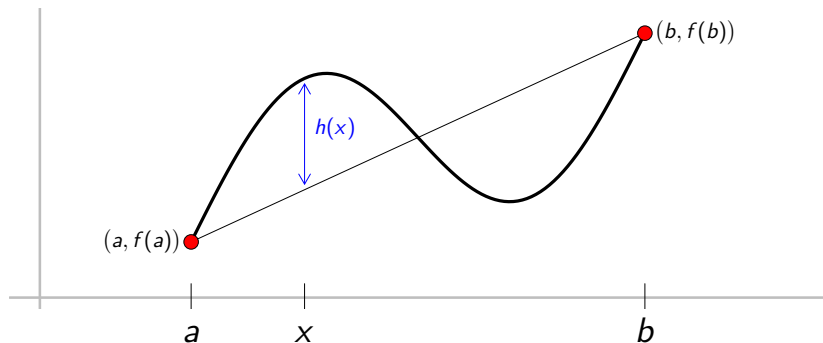
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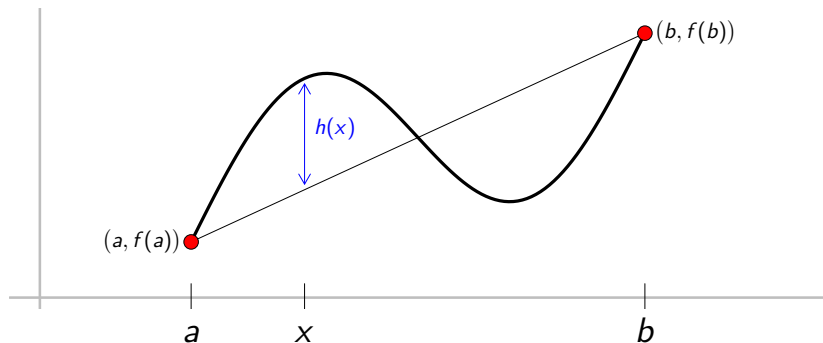
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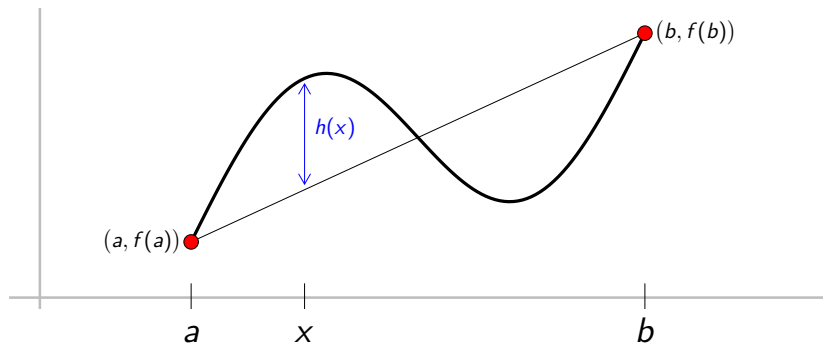
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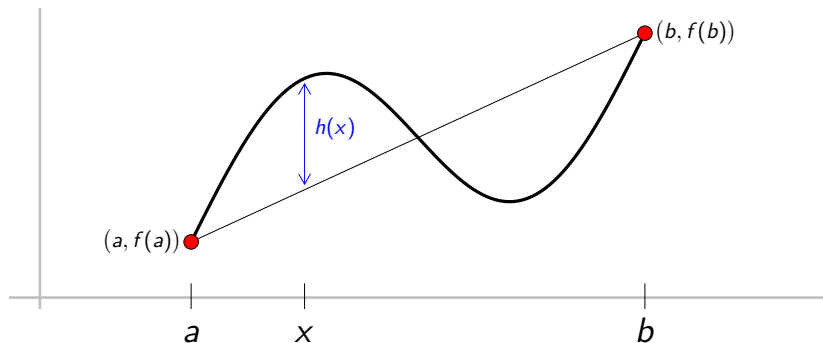
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Apply Rolle's theorem to

$$h(x) = f(x) - \left[f(a) \right]$$

The Mean Value Theorem

Idea for proof:



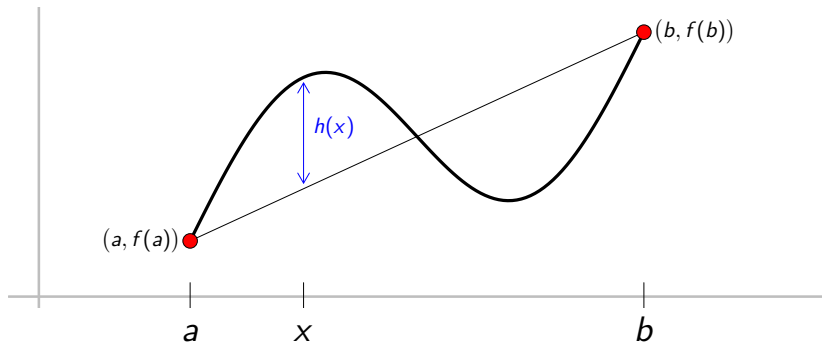
Proof.

Apply Rolle's theorem to

$$h(x) = f(x) - \left[f(a) + \right.$$

The Mean Value Theorem

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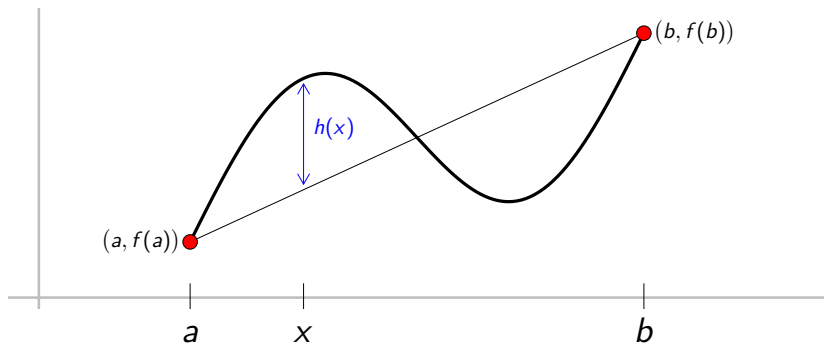
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Apply Rolle's theorem to

$$h(x) = f(x) - \left[f(a) + \left(\frac{f(b) - f(a)}{b - a} \right) (x - a) \right]$$

The Mean Value Theorem

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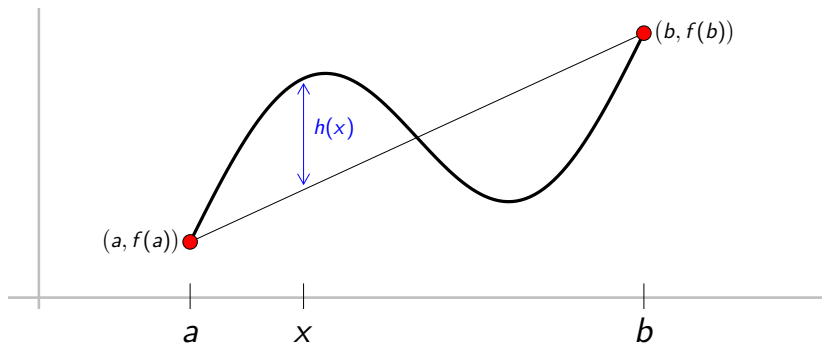
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The Mean Value Theorem

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Example

$f'(x) > 0$ on an interval $I \implies f$ strictly increasing on I .

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Suppose $x_1, x_2 \in I$

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Suppose $x_1, x_2 \in I$ and $x_1 < x_2$.

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Suppose $x_1, x_2 \in I$ and $x_1 < x_2$. We must show

The Mean Value Theorem

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Suppose $x_1, x_2 \in I$ and $x_1 < x_2$. We must show $f(x_1) < f(x_2)$.

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The Mean Value Theorem

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Since $f'(x)$ exists for all $x \in I$,

The Mean Value Theorem

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Suppose $x_1, x_2 \in I$ and $x_1 < x_2$. We must show $f(x_1) < f(x_2)$.

Since $f'(x)$ exists for all $x \in I$, f is certainly differentiable

The Mean Value Theorem

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The Mean Value Theorem

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$\therefore f(x_2) - f(x_1) > 0$,

The Mean Value Theorem

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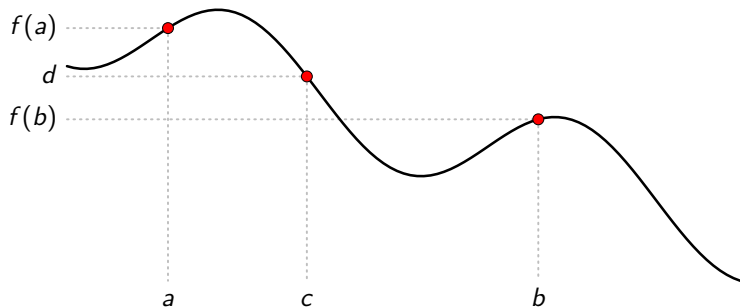
$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(x_*).$$

But $x_2 - x_1 > 0$ and since $x_* \in I$, we know $f'(x_*) > 0$.

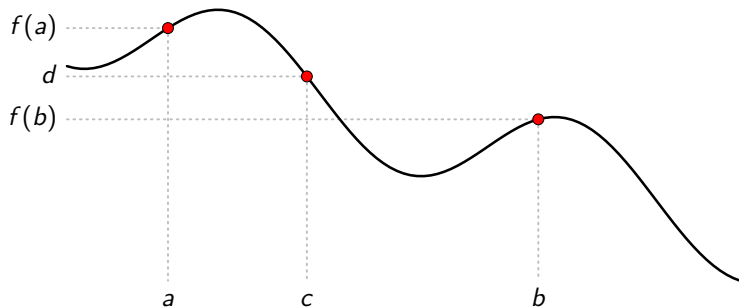
$\therefore f(x_2) - f(x_1) > 0$, i.e., $f(x_1) < f(x_2)$. □

REMINDER: Intermediate Value Property

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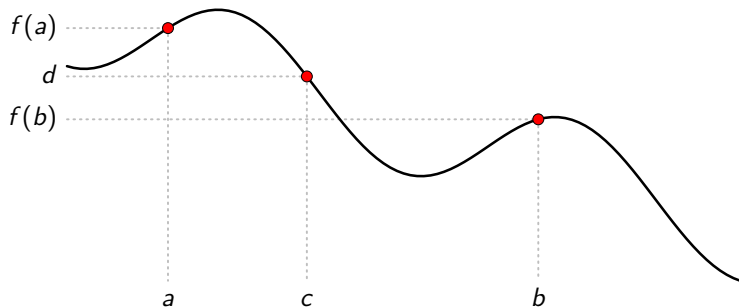


REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

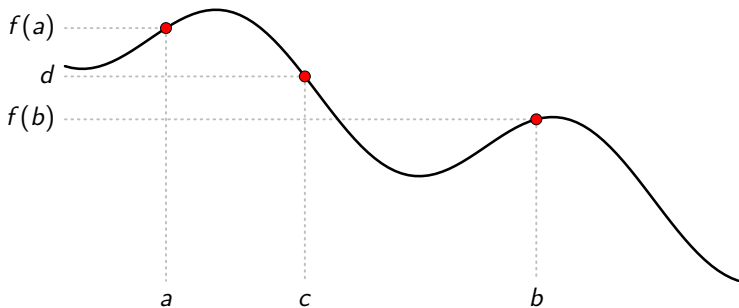
REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f

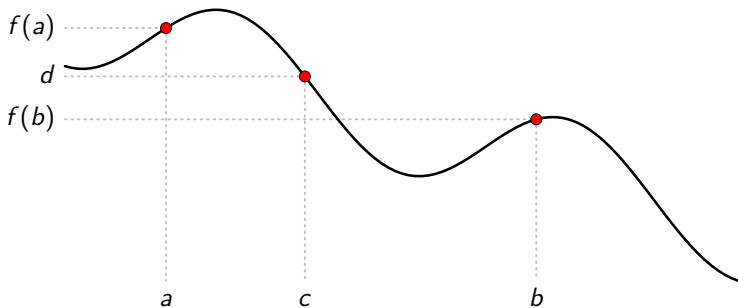
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Definition (Intermediate Value Property (IVP))

A function f defined on

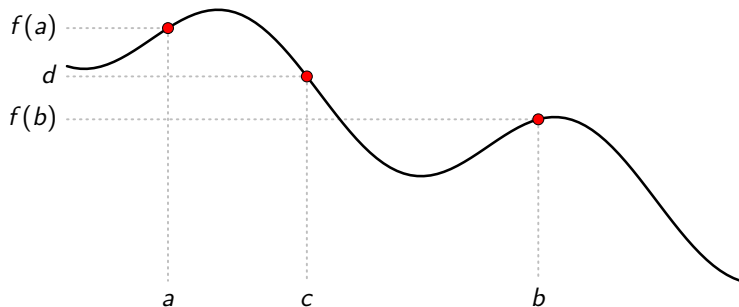
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Definition (Intermediate Value Property (IVP))

A function f defined on an interval I

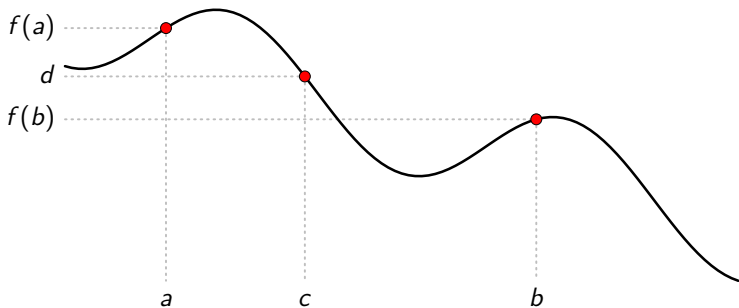
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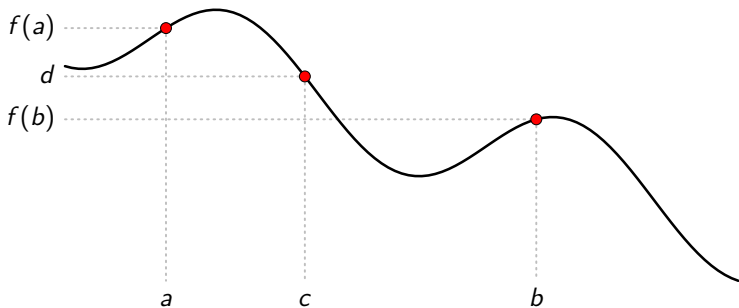
REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f defined on an interval I is said to have the *intermediate value property (IVP)*

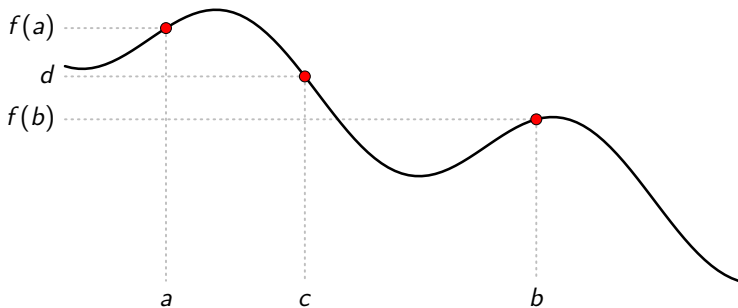
REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f defined on an interval I is said to have the *intermediate value property (IVP)* on I

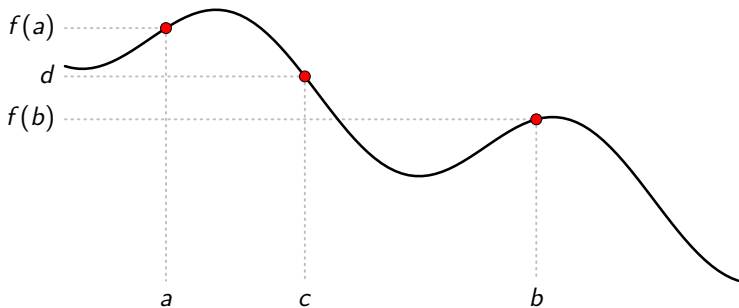
REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f defined on an interval I is said to have the **intermediate value property (IVP)** on I iff

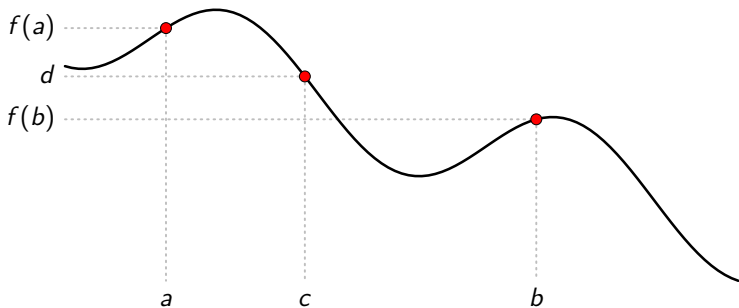
REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f defined on an interval I is said to have the **intermediate value property (IVP)** on I iff for each $a, b \in I$

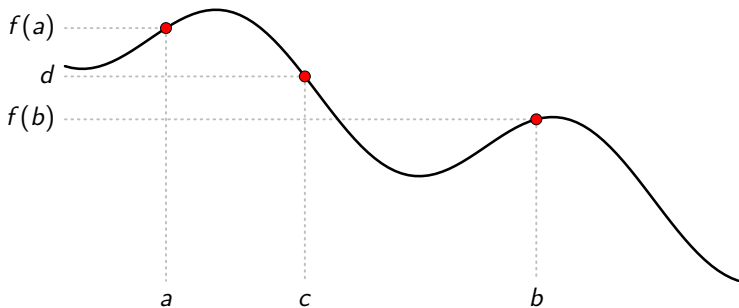
REMINDER: Intermediate Value Property



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A function f defined on an interval I is said to have the **intermediate value property (IVP)** on I iff for each $a, b \in I$ with $f(a) \neq f(b)$

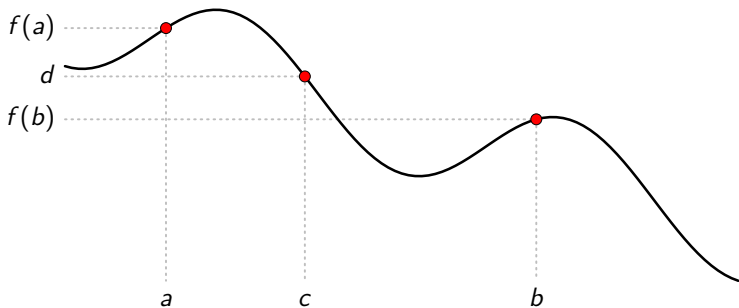
REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f defined on an interval I is said to have the **intermediate value property (IVP)** on I iff for each $a, b \in I$ with $f(a) \neq f(b)$, and for each d

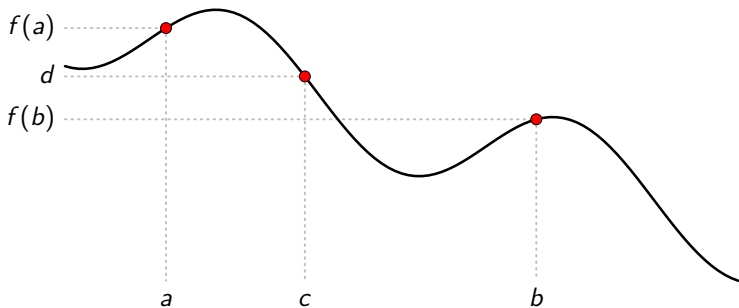
REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f defined on an interval I is said to have the **intermediate value property (IVP)** on I iff for each $a, b \in I$ with $f(a) \neq f(b)$, and for each d between $f(a)$ and $f(b)$,

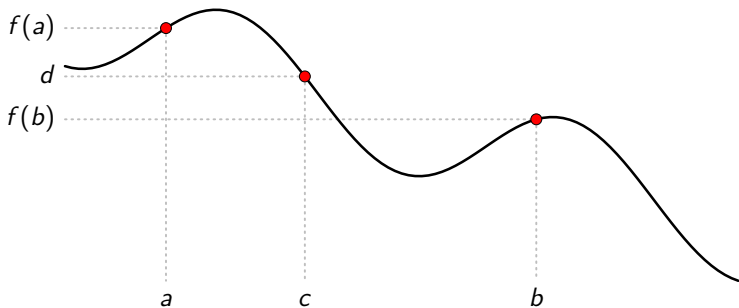
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A function f defined on an interval I is said to have the **intermediate value property (IVP)** on I iff for each $a, b \in I$ with $f(a) \neq f(b)$, and for each d between $f(a)$ and $f(b)$, there exists c

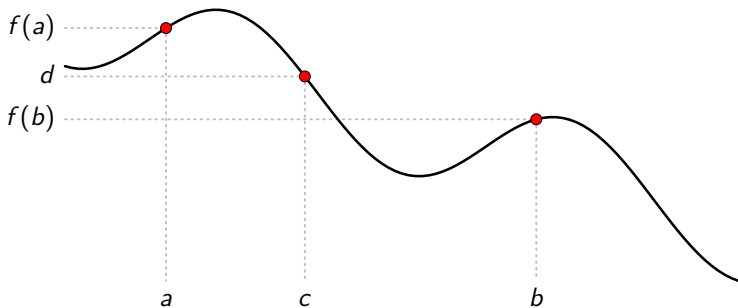
REMINDER: Intermediate Value Property



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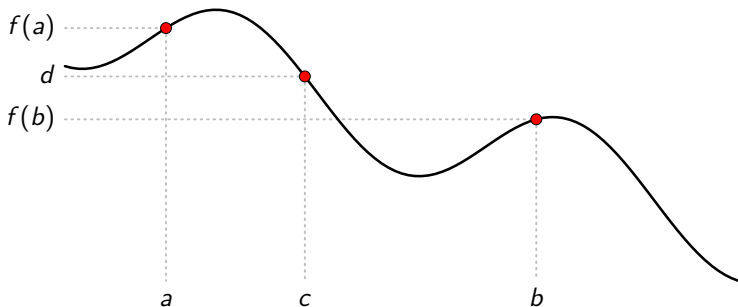
REMINDER: Intermediate Value Property



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REMINDER: Intermediate Value Property



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REMINDER: Intermediate Value Property

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Theorem (Intermediate Value Theorem (IVT))

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Theorem (Intermediate Value Theorem (IVT))

If f is continuous

REMINDER: Intermediate Value Property

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If f is continuous on an interval I

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Theorem (Intermediate Value Theorem (IVT))

If f is continuous on an interval I then

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If f is continuous on an interval I then f has

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Theorem (Intermediate Value Theorem (IVT))

*If f is continuous on an interval I then f has the **intermediate value property (IVP)***

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Note:

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Theorem (Intermediate Value Theorem (IVT))

*If f is continuous on an interval I then f has the **intermediate value property (IVP)** on I .*

Note: The interval I in the statement of the IVT

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

*If f is continuous on an interval I then f has the **intermediate value property (IVP)** on I .*

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REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

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REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

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Unlike the **extreme value theorem**,

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

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Note: The interval I in the statement of the IVT does not have to be closed and it does not have to be bounded.

Unlike the **extreme value theorem**, the IVT is not a theorem about functions defined on closed and bounded intervals.

Poll

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 3A03**
- Click on **Take Class Poll**
- Fill in poll **Derivatives: IVP and derivatives**
- .

Intermediate Value Property

Intermediate Value Property

Question:

Intermediate Value Property

Question: If a function has the IVP on an interval I , must it be continuous on I ?

Intermediate Value Property

Question: If a function has the IVP on an interval I , must it be continuous on I ?

Example

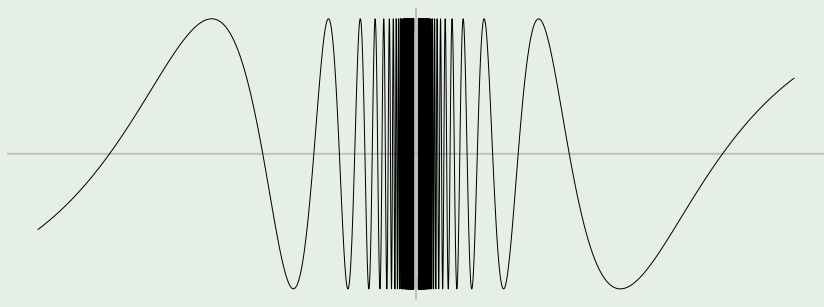
$$f(x) = \begin{cases} \sin \frac{1}{x} & x \neq 0, \\ 0 & x = 0. \end{cases}$$

Intermediate Value Property

Question: If a function has the **IVP** on an interval I , must it be **continuous** on I ?

Example

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Intermediate value property for derivatives

Intermediate value property for derivatives

Theorem (Darboux's Theorem: IVP for derivatives)

Intermediate value property for derivatives

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If f is differentiable

Intermediate value property for derivatives

Theorem (Darboux's Theorem: IVP for derivatives)

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Intermediate value property for derivatives

Theorem (Darboux's Theorem: IVP for derivatives)

If f is differentiable on an interval I then

Intermediate value property for derivatives

Theorem (Darboux's Theorem: IVP for derivatives)

If f is differentiable on an interval I then its derivative

Intermediate value property for derivatives

Theorem (Darboux's Theorem: IVP for derivatives)

If f is differentiable on an interval I then its derivative f'

Intermediate value property for derivatives

Theorem (Darboux's Theorem: IVP for derivatives)

If f is differentiable on an interval I then its derivative f' has the intermediate value property

Intermediate value property for derivatives

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Intermediate value property for derivatives

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Intermediate value property for derivatives

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Notes:

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Intermediate value property for derivatives

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Intermediate value property for derivatives

Theorem (Darboux's Theorem: IVP for derivatives)

*If f is differentiable on an interval I then its derivative f' has the **intermediate value property** on I .*

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- It is f' , not f , that is claimed to have the **intermediate value property** in Darboux's theorem. This theorem does not follow from the standard **intermediate value theorem**

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Intermediate value property for derivatives

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Proof of Darboux's Theorem.

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Consider $a, b \in I$ with $a < b$.

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Suppose first that $f'(a) < 0 < f'(b)$. We will show $\exists x \in (a, b)$ such that $f'(x) = 0$.

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Therefore, $x \in (a, b)$. But f is differentiable everywhere in (a, b) , so, by the **theorem on the derivative at local extrema**, we must have $f'(x) = 0$. Now suppose more generally that $f'(a) < K < f'(b)$.

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Proof of Darboux's Theorem.

Consider $a, b \in I$ with $a < b$.

Suppose first that $f'(a) < 0 < f'(b)$. We will show $\exists x \in (a, b)$ such that $f'(x) = 0$. Since f' exists on $[a, b]$, we must have f continuous on $[a, b]$, so the **Extreme Value Theorem** implies that f attains its minimum at some point $x \in [a, b]$. This minimum point cannot be an endpoint of $[a, b]$ ($x \neq a$ because $f'(a) < 0$ and $x \neq b$ because $f'(b) > 0$).

Therefore, $x \in (a, b)$. But f is differentiable everywhere in (a, b) , so, by the **theorem on the derivative at local extrema**, we must have $f'(x) = 0$.

Now suppose more generally that $f'(a) < K < f'(b)$. Let

$g(x) = f(x) - Kx$. Then g is differentiable on I and $g'(x) = f'(x) - K$ for all $x \in I$. In addition, $g'(a) = f'(a) - K < 0$ and

$g'(b) = f'(b) - K > 0$, so by the argument above, $\exists x \in (a, b)$ such that $g'(x) = 0$, i.e., $f'(x) - K = 0$, i.e., $f'(x) = K$.

The case $f'(a) > K > f'(b)$ is similar. □

Intermediate value property for derivatives

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Example ($f'(x) \neq 0 \forall x \in I \implies f \nearrow$ or $\searrow$ on I)

If f is differentiable on an interval I and $f'(x) \neq 0$ for all $x \in I$ then f is either increasing or decreasing on the entire interval I .

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Then, from [Darboux's theorem](#), $\exists c \in I$ such that $f'(c) = 0$.

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i.e., either f is increasing on I or decreasing on I . □

The Derivative of an Inverse

The Derivative of an Inverse

Example (Sufficient condition for *non*-differentiable inverse)

Suppose f is continuous and one-to-one on an interval I . If $x \in I$ and $f'(x) = 0$ then f^{-1} is not differentiable at $y = f(x)$.

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The Derivative of an Inverse

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Theorem (Inverse function theorem)

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If f is differentiable on an interval I

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- 3** $(f^{-1})'(f(x)) = \frac{1}{f'(x)}$ for all $x \in I$,
i.e., $(f^{-1})'(y) = \frac{1}{f'((f^{-1}(y)))}$ for all $y \in J$.

(TBB Theorem 7.32, p. 445)

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- In fact, f continuous and either $\nearrow$ or $\searrow$ on $I \implies f^{-1}$ continuous on $J = f(I)$. (more generally, cf. Invariance of Domain thm)